Gulati 2020 breast cancer — analysis summary

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

7/8 samples (≥54 cells) intermix on one continuous manifold with no per-patient islands (radius-graph: 0 detached; iLISI median 2.47→4.79 of 7); both lobes (Low-cycling 513 / High-cycling 305) carry the trajectory.

Gate 2 — module geography → NO-GO

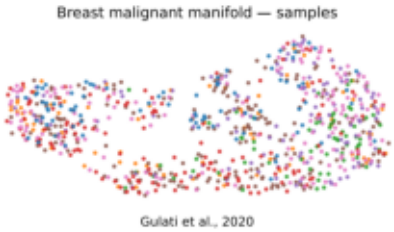
Developmental geography is ABSENT in raw per-cell scores (Gast↔pSom = −0.02) and appears only after graph-smoothing. A permutation null (smoothing random scores on the same graph) reproduces it: smoothed −0.77 within null 95% CI ±0.7; the defining terminal-Crest gradient (ρ=+0.73) is non-significant (p=0.14). The programs are present but not organised into a developmental axis — pattern absent.

Gate 3 — cross-repression → NA

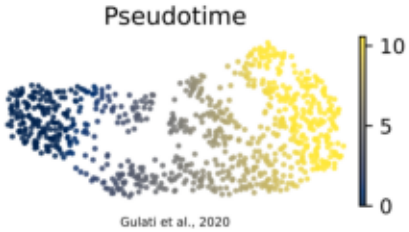
Not reached — Gate 2 NO-GO halts the gate chain; branching-only, and there is no bifurcation or Mixed-type hub to test.

STEP FIGURES

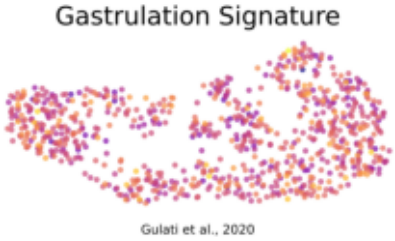
Manifold — samples



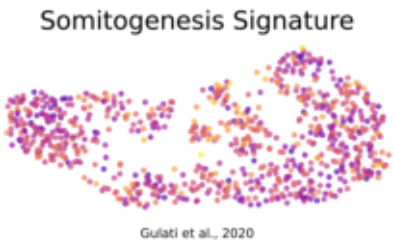
Pseudotime



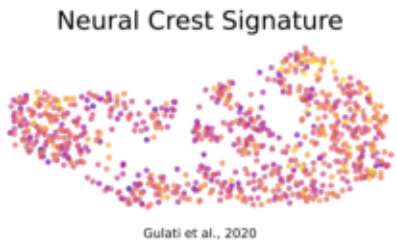
Gastrulation



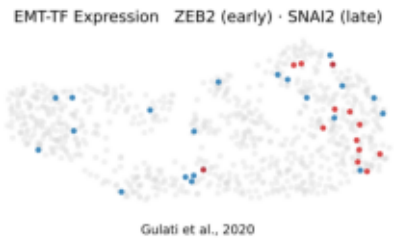
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.

Breast cancer (Gulati et al., 2020), SmartSeq2 — 15 samples / 1,902 cells (master scope); malignant only: 861 load, 818 across 7 samples (≥54) enter preprocessing; whole malignant manifold (no lobe scoping). A trajectory is fit, but the module pattern is absent (Gate 2 NO-GO).

Verdict.

Pattern does NOT reproduce — Gate 1 GO, Gate 2 NO-GO, Gate 3 NA. Gate 1 holds (818 cells intermix on one continuous manifold), but the embryonic-module geography is absent at single-cell resolution: raw Gast↔pSom = −0.02 (no cross-repression), and the smoothed structure (−0.77; terminal Crest ρ=+0.73) is not distinguishable from a graph-smoothing artifact — a permutation null on this 818-cell SmartSeq2 graph gives 95% CI ±0.7, terminal-Crest non-significant (p=0.14). EMT-TFs (unconditional step) ZEB2 (early) → SNAI2 (late).

Parameter choices

Min cells/sample	≥54 (7/8; gap 68→43)	**Scope**	whole manifold (no scoping)
Root node	tip 2 (pSom/Gast-high)	**Integration**	Harmony θ=2; PCA20; spread=1.3
Trajectory	scFates curve N=20, λ=0.3, μ=100 (PPT forks unstable 2→8)	**Compartments**	Low-cyc 513 / High-cyc 305 (cell cycle orthogonal, |r|≤0.03)
EMT-TFs	ZEB2 (early) → SNAI2 (late); mesenchymal factors lean crest-terminal		